

German physicist Max Planck's quantum theory refuted the ideas of classical physics by showing that energy is emitted not continuously, but in fixed packets, or "quanta." It fundamentally altered the way scientists interpreted the subatomic world.

MILESTONES

HEAT THEORY WORK

Advances his study of heat theory from 1885 to 1889, while associate professor of physics at Kiel University.

STUDIES RADIATION

Investigates black body radiation and calculates the vibration frequencies of atoms during the 1890s.

MAKES BREAKTHROUGH

Presents his radiation distribution law in 1900, introducing the concept of energy quanta to physics.

HELPS EINSTEIN

In 1905, is one of the first prominent physicists to publicly support Einstein's Theory of Relativity.

AWARDED NOBEL PRIZE

Receives Nobel Prize in Physics in 1919 for his groundbreaking discovery of energy quanta.

Planck popularized the work of his friend Albert Einstein in Germany and created a new professorship specifically for him in 1914, at the University of Berlin.

Born in Kiel, northern Germany, Max Planck was the youngest of six children. When he was 9, the family moved to Munich, and Planck attended the Maximilian gymnasium. There, he showed an aptitude for mechanics, mathematics, and music, and after one tutor sparked his interest in physics, he chose to study the subject in college. At 21, he received a doctoral degree from the University of Munich, submitting a thesis on the second law of thermodynamics (how heat moves). In 1889, he became professor of theoretical physics at the University of Berlin, where he remained until he retired in 1926.

Black body radiation

In the 1890s, physicists were struggling to explain the absorption and emission of light and became preoccupied with "black body" radiation. In 1859, German physicist Gustav Kirchhoff had defined a black body as a hypothetical body that absorbs all the electromagnetic radiation, including light, that falls on it. When it is heated, a black body radiates energy in the form of electromagnetic waves with a range of wavelengths, including visible, ultraviolet, and infrared light.

However, during their experiments, physicists had noted that the wavelengths radiated by hot objects were not the same as those predicted by classical theories of thermodynamics. In 1894, while in Berlin, Planck turned his attention to black body radiation, beginning a quest to match theory with observation.

